

AMANDA ROYKA

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APPOINTMENTS

Harvard Society of Fellows <i>Junior Fellow, Psychology</i>	2026 –
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EDUCATION

Yale University <i>PhD., Psychology</i> <i>Advisors:</i> Julian Jara-Ettinger & Laurie Santos	2026
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University of Edinburgh <i>Master's in Linguistics by Research</i> <i>Advisors:</i> Simon Kirby & Marieke Schouwstra <i>Dissertation Topic:</i> An examination of the features of symbolic communicative gestures and how those features may serve to signal communicative intent to observers	2019-2020
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Queen Mary University of London <i>Master's in Ecology and Evolutionary Biology with Distinction</i> <i>Advisor:</i> Lars Chittka <i>Dissertation Topic:</i> An investigation of cumulative cultural transmission in large earth bumblebees	2018 – 2019
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Yale University <i>BS in Cognitive Science, Summa Cum Laude</i> <i>Advisors:</i> Laurie Santos & Angie Johnston <i>Thesis:</i> A comparative study of metacognitive information-seeking in dingoes and domesticated dogs in order to examine the relationship between the cognitive representations underlying a sensitivity to mental states	2014 – 2018
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AWARDS AND HONORS

Junior Fellow, Harvard Society of Fellows	2026
Miller Postdoctoral Fellowship, \$270,000 [declined]	2026
Yale's James B. Grossman Award for outstanding dissertation work in psychology	2026
Besample Dissertation Grant	2026
William James Prize Finalist at the Annual Meeting of the Society for Philosophy and Psychology	2025
Marr Prize for best student paper at the Annual Meeting of the Cognitive Science Society	2024
Diverse Intelligences Summer Institute Project Accelerator Grant, \$3,000	2024
Yale Council of Latin American and Iberian Studies, \$500	2023
Yale Ender's Fellowship Grant, \$2,000	2022
Marshall Scholar	2018 – 2020
Rhodes Scholarship Finalist	2018
Yale Angiers Prize for best undergraduate thesis in psychology	2018

Publications

Berke, M*, Horschler, D*, **Royka, A.**, Santos, L., & Jara-Ettinger, J. (*accepted*). What Primates Know About Other Minds and When They Use It: A Computational Approach to Comparative Theory of Mind. *Cognition*.

Royka, A.*, Horschler, D. J.*, Bargmann, W., & Santos, L. R. (2026). Exploring the evolutionary roots of theory of mind: Primate errors on false belief tasks reveal representational limits. *Cognition*, 270, 106400.

Royka, A., & Santos, L. R. (2025). Informing does not require attributing ignorance. *Proceedings of the National Academy of Sciences of the United States of America*, 122(20), e2504155122.

Royka, A., Choi, K., Schoenberger, R., & Jara-Ettinger, J. (2025). Interjections as Tools for Sharing Mental States. *47th Annual Meeting of the Cognitive Science Society*.

Rane, S., Kirkman, C., **Royka, A.**, Todd, G., Law, R., Gates Foster, J., & Cartmill, E. (in press). Principles of Animal Cognition for LLM Evaluations: A Case Study on Transitive Inference. Accepted as a position paper. ICML 2025.

Muchovej, J., **Royka, A.**, Lee, S., & Jara-Ettinger, J. (2025). GPT-4o Lacks Core Features of Theory of Mind. *47th Annual Meeting of the Cognitive Science Society*.

Bridges, A., **Royka, A.**, Wilson, T., Lockwood, C., Richter, J., Juusola, M., & Chittka, L. (2024). Bumblebees socially learn behaviour too complex to innovate alone. *Nature*.

Royka, A.*, Horschler, D*, Bargmann, W., & Santos, L. (2024). Probing Nonhuman Primate Errors on False Belief Tasks to Explore the Evolutionary Roots of Theory of Mind. *46th Annual Meeting of the Cognitive Science Society*.

Royka, A., Heuser, G., Schwoustra, M., Kirby, S., & Jara-Ettinger, J. (2024). Emblems and Improvised Gestures are Structures to Guide their Own Detection. *46th Annual Meeting of the Cognitive Science Society*.

Royka, A., Torok, G., & Jara-Ettinger, J. (2023). Guiding Inference: Signaling intentions using efficient action. *45th Annual Meeting of the Cognitive Science Society*.

Royka, A., Chen, A., Aboody, R., Huanca, T., & Jara-Ettinger, J. (2022). People infer communicative action through an expectation for efficient communication. *Nature Communications*. 13(1), 1-13.

Royka, A., & Santos, L. R. (2022) Theory of Mind in the Wild. *Current Opinions in Behavioral Sciences*.

Royka, A. L., & Jara-Ettinger, J. (2021). Commentary: Ignorance Matters. *Behavioral and Brain Sciences*.

McAuliffe, K., Drayton, L. A., **Royka, A.**, Aellen, M., Santos, L. R., & Bshary, R. (2021). Cleaner fish are sensitive to what their partners can and cannot see. *Communications Biology*.

Royka, A., Schouwstra, M., Kirby, S., & Jara-Ettinger, J. (2021). I Know You Know I'm Signaling: Novel gestures are designed to guide observers' inferences about communicative goals. *43th Annual Meeting of the Cognitive Science Society*.

Royka, A. L., Johnston, A. M., & Santos, L. R. (2020). Metacognition in canids: A comparison of dogs (*Canis familiaris*) and dingoes (*Canis dingo*). *Journal of Comparative Psychology*, 134(3), 303.

Johnson, S. G. B., **Royka, A.**, McNally, P., & Keil, F. (2019). When is science considered interesting and important?. *41st Annual Meeting of the Cognitive Science Society*.

Royka, A., Aboody, R., & Jara-Ettinger, J. (2018). Movement as a message: inferring communicative intent from actions. *40th Annual Meeting of the Cognitive Science Society*.

Manuscripts in Preparation

Royka, A., Muchovej, J., Heuser, G., & Jara-Ettinger, J. (*submitted*). Emblems are Structures to Guide their Own Detection.

Royka, A., Schwoustra, M., Kirby, S., & Jara-Ettinger, J. (*in prep*). People Improvise Gestures to Guide Observers' Communicative Inferences.

Royka, A.*., Suwal, U*., & Jara-Ettinger, J. (*in prep*). Variability in the Speed, Accuracy, and Effort of Mental-State Inferences in Theory of Mind.

Muchovej, J., **Royka, A.**, Lee, S., & Jara-Ettinger, J. (*in prep*). GPT-4o Lacks Core Features of Theory of Mind.

CONFERENCE PRESENTATIONS AND INVITED TALKS

Presentation: **Royka, A.**, Townrow, L., Baker, A., Berke, A. & Santos, L. (2026). Monkeys fail intuitive physics inference tasks. Annual Meeting of the Society for Philosophy and Psychology.

Invited Talk: What we are missing about Theory of Mind (2026). Consortium for Interacting Minds, Dartmouth College.

Invited Talk: Sketching the System: Mapping Theory of Mind across species (2025). Social & Cognitive Origins Group, Johns Hopkins University.

Invited Talk: Sketching the System: Mapping Theory of Mind across species (2025). Max Planck Institute for Evolutionary Anthropology, Department of Comparative Cultural Psychology. Leipzig, Germany.

Presentation: **Royka, A.**, Heuser, G., Schouwstra, M., Kirby, S. & Jara-Ettinger, J. (2025). Emblems and Improvised Gestures are Structured to Guide their Own Detection. The International Conference on Gesture Studies.

Poster: **Royka, A.**, Choi, K., Schoenberger, R. & Jara-Ettinger, J. (2025). Interjections as Tools for Sharing Mental States. 47th Annual Meeting of the Cognitive Science Society.

Presentation: **Royka, A.**, Choi, K., Schoenberger, R. & Jara-Ettinger, J. (2025). Interjections as Tools for Sharing Mental States. Annual Meeting of the Society for Philosophy and Psychology.

Poster: **Royka, A.**, Horschler, D., Bargmann, W., & Santos, L. (2024). Probing Nonhuman Primate Errors on False Belief Tasks to Explore the Evolutionary Roots of Theory of Mind. Dubrovnik Cognitive Science Conference.

Presentation: Muchovej, J., Lee, J., **Royka, A.**, & Jara-Ettinger, J. (2025). Large Language Models Show No Coherence Across Different Theory of Mind Tasks: Evidence From GPT-4o. Advancing Artificial Intelligence through Theory of Mind (ToM4AI): Bridging Human Cognition and Artificial Intelligence Workshop @AAAI 2025.

Poster: Rane, S., Kirkman, C., **Royka, A.**, Todd, G., Law, R., Gates, J., & Cartmill, E. (2024). Principles of Animal Cognition for LLM Evaluations: A Case Study on Transitive Inference. Behavioral Machine Learning Workshop @NeurIPS 2024.

Presentation: **Royka, A.**, Horschler, D., Bargmann, W., & Santos, L. (2024). Probing Nonhuman Primate Errors on False Belief Tasks to Explore the Evolutionary Roots of Theory of Mind. 46th Annual Meeting of the Cognitive Science Society.

Presentation: **Royka, A.**, Heuser, G., Schwoustra, M., Kirby, S., & Jara-Ettinger, J. (2024). Emblems and Improvised Gestures are Structures to Guide their Own Detection. 46th Annual Meeting of the Cognitive Science Society.

Invited Talk: Gesture as Mindsharing (2024). ConCats, New York University.

Guest Lecture: 20 Years of Theory of Mind Research on Cayo Santiago (2024). Trinity College.

Invited Talk: Shaped to Communicate: Gesture as a form of goal inference (2023). Consortium for Interacting Minds, Dartmouth College.

Poster: **Royka, A.**, Torok, G., & Jara-Ettinger, J. (2023). Guiding Inference: Signaling intentions using efficient action. Computational Summer School on Modeling Social and Collective Behavior.

Poster: **Royka, A.**, Torok, G., & Jara-Ettinger, J. (2023). Guiding Inference: Signaling intentions using efficient action. 45th Annual Meeting of the Cognitive Science Society. [Virtual]

Presentation: **Royka, A.**, Schwoustra, M., Kirby, S., & Jara-Ettinger, J. (2021). I Know You Know I'm Signaling: Novel gestures are designed to guide observers' inferences about communicative goals. Society for Philosophy and Psychology.

Presentation: **Royka, A.**, Schwoustra, M., Kirby, S., & Jara-Ettinger, J. (2021). I Know You Know I'm Signaling: Novel gestures are designed to guide observers' inferences about communicative goals. 43rd Annual Meeting of the Cognitive Science Society.

Presentation: **Royka, A.**, Aboody, R., & Jara-Ettinger, J. (2018). Movement as a message: inferring communicative intent from actions. 40th Annual Meeting of the Cognitive Science Society.

Poster: **Royka, A.L.**, Johnston, A.M. & Santos, L. R. (2017). Metacognition across domestication: A comparison of dogs and dingoes. Poster Presentation, North American International Canine Science Conference.

TEACHING

TA, Introduction to Psychology	Spring 2023
TA, Introduction to Cognitive Science	Fall 2022
TA, Perspectives on Human Nature	Spring 2022
TA, Introduction to Psychology	Fall 2021

ADDITIONAL TEACHING TRAINING

Mental Health in the Classroom	Fall 2025
Inclusive Syllabus Design	Spring 2025
Antiracist Pedagogy	Spring 2024
Using Group Work Effectively	Spring 2024
Fundamentals of Equitable Teaching	Fall 2023

SERVICE

Ad-hoc reviewing for Science, Cognition, PNAS, BBS, and OpenMind	
Assistant Director of Undergraduate Studies for Psychology and Cognitive Science	Fall 2025 – Spring 2026
Mentor for Yale's Sneak Peek Program, assisting URM students applying to grad school	Fall 2022 – Fall 2026
Residential College Graduate Affiliate	Spring 2022 – Fall 2026
Organized department panel on applying to non-academic careers	Spring 2025
Organized department panel on applying to academic jobs	Spring 2024
Organized a department workshop on inclusive pedagogy	Spring 2023
Student representative on department colloquium committee	Spring 2021 – Spring 2023
Student representative on cognitive-developmental current works speaker series committee	Summer 2021 – Spring 2022
Student representative on the developmental area space planning committee	Fall 2020 – Spring 2021